

DAG Pipeline Declared by pipeline.yaml |

Single-project pipelines read `infrastructure/core/pipeline/pipeline.yaml`. `scripts/execute_pipeline.py` expands the declarative DAG, applies tag filters (`--core-only` skips llm stages), checkpoints between nodes, then dispatches numbered scripts (`scripts/NN_*.py`) or builtin methods (`_run_clean_outputs`).

The **default YAML graph contains ten named stages** (plus telemetry configuration metadata):

1. **Clean Output Directories** — wipes prior `projects/<name>/output/` + delivered `output/<name>/` paths so stale PDFs cannot satisfy validation.
2. **Environment Setup** (`00_setup_environment.py`) — Python/uv probing, toolchain discovery, scaffolding directories, `PYTHONPATH` wiring.
3. **Infrastructure Tests** (`01_run_tests.py --infra-only`) — tests/ suite with infra coverage thresholds (60 %).

DAG Pipeline Declared by pipeline.yaml II

4. **Project Tests** (01_run_tests.py --project-only) — per-project suites with 90 % coverage mandate.
5. **Project Analysis** (02_run_analysis.py) — lexicographically ordered projects/<name>/scripts/*.py, each a thin orchestrator (src/ does real work).
6. **PDF Rendering** (03_render_pdf.py) — Pandoc → XeLaTeX loop, bibliography assembly, injected variables from Stage 02 artefacts.
7. **Output Validation** (04_validate_output.py) — PDF structure, manifests, Markdown hygiene.
8. **LLM Scientific Review** (06_llm_review.py --reviews-only; tags: llm) — executive + quality critiques via local Ollama; allow_skip: true.
9. **LLM Translations** (06_llm_review.py --translations-only; tags llm, same dependency edges) — multilingual abstract expansion.

DAG Pipeline Declared by `pipeline.yaml` III

10. **Copy Outputs** (`05_copy_outputs.py`) — reproducible snapshots into canonical `output/<project>/`.

Two LLM nodes intentionally share one script module with orthogonal CLI switches; both depend only on validation so they can parallelize logically while remaining optional.

Executive reporting (`scripts/07_generate_executive_report.py`) is **not** a YAML node inside the single-project executor. `--all-projects / execute_multi_project.py` invokes it once after iterating projects, consolidating cross-project KPIs dashboards.

Topological order therefore differs slightly from lexical script numbering (e.g., `copy` executes after `validation` even though script 05 precedes 06 lexically).

DAG Pipeline Declared by `pipeline.yaml` IV

Stage Highlights

Infrastructure vs project tests. Splitting `pytest` invocations isolates flaky infra regressions (`MAX_TEST_FAILURES` knobs) from zero-tolerance gates on domain code (`max_project_test_failures` default 0 declared in YAML front-matter/testing blocks).

Stage 02 illustration. The analysis stage is deliberately concrete rather than a hypothetical diagram factory: each canonical project ships real behaviour at this node. `template_autoresearch_project` runs readiness validation; the archived `template_search_project` merges remote literature JSON, generates scripted figures (`y_generate_search_figures.py`), and writes manifests; `template_code_project` emits optimization plots; and `template_prose_project` triggers structural validation scaffolding. The pipeline shape is identical across all four—only the Stage 02 payload differs—which is exactly what lets one orchestrator serve heterogeneous research domains.

DAG Pipeline Declared by pipeline.yaml V

Interactive Orchestration

run.sh

Thin wrapper invoking `python -m infrastructure.orchestration`. Offers:

- ▶ per-project staged execution,
- ▶ chained digits (234 shorthand),
- ▶ multi-project grid (a-d presets),
- ▶ graceful quit / resume parity with `scripts/README.md`.

Selecting **d alone** after a passing multi-project run exits immediately once summaries print—avoiding repetitive menu redraw.

secure_run.sh

Executes Python secure path: standard pipeline artefact reproduction **then** invokes `run_secure_pipeline` for steganographic PDF hardening (`infrastructure.steganography`). Original PDFs stay immutable; hardened companions carry QR overlays plus hash manifests sidecars.